dynamic Aspect of Ecology," it was about the relationship between food chains and the changes over time in a community of species.

Lindeman had spent five years studying the life forms in an aging lake at Cedar Creek Bog, Minnesota, and was especially interested in the changes in the lake as, year by year, aquatic habitat gradually gave way to land. He received his Ph.D., but his paper was initially rejected by *Ecology*, for being too theoretical.

Lindeman had painstakingly sampled everything in the lake, from the aquatic plants and microscopic algal plankton to the worms, insects, crustaceans, and fish that fed upon one another and depended on each other for their existence. He stressed that the community of organisms could not be properly understood on its own; instead, it must be examined in the context of its wider surroundings. The living (biotic) organisms and the nonliving (abiotic) components (air, water, soil minerals) were linked together by nutrient cycles and energy flows. This entire system—the ecosystem—was the central ecological unit.

